

RaceDay System Documentation

API Endpoint Plan

This section specifies every REST API endpoint for RaceDay. All routes are prefixed with `/api`. Role values: Public (no login), Participant, Organiser, Any (logged-in).

2.1 Authentication

Method	Route	Description	Role Required
POST	<code>/api/auth/register</code>	Register a new user	Public
POST	<code>/api/auth/login</code>	Authenticate and return JWT	Public
POST	<code>/api/auth/logout</code>	Invalidate session	Any (logged-in)
GET	<code>/api/auth/refresh-token</code>	Issue new access token	Any (logged-in)

2.2 User Profile

Method	Route	Description	Role Required
GET	<code>/api/users/{id}</code>	Get user basic info	Any (logged-in)
PUT	<code>/api/users/{id}</code>	Update own account	Any (owner only)
GET	<code>/api/users/{id}/profile</code>	Get role-specific profile	Any (owner only)

2.3 Events

Method	Route	Description	Role Required
GET	<code>/api/events</code>	List all events	Public
GET	<code>/api/events/{id}</code>	Get event details	Public
POST	<code>/api/events</code>	Create new event	Organiser
PUT	<code>/api/events/{id}</code>	Edit event	Organiser
DELETE	<code>/api/events/{id}</code>	Delete event	Organiser

2.4 Event Categories

Method	Route	Description	Role Required
GET	<code>/api/events/{eventId}/categories</code>	List categories	Public
POST	<code>/api/events/{eventId}/categories</code>	Add category	Organiser
PUT	<code>/api/categories/{id}</code>	Edit category	Organiser

Method	Route	Description	Role Required
DELETE	/api/categories/{id}	Delete category	Organiser

2.5 Event Enrolments

Method	Route	Description	Role Required
GET	/api/enrolments	List my enrolments	Participant
POST	/api/enrolments	Enrol in event	Participant
GET	/api/events/{id}/enrolments	List event enrolments	Organiser
DELETE	/api/enrolments/{id}	Withdraw from event	Participant

2.6 Results / Performance

Method	Route	Description	Role Required
GET	/api/results/{enrolmentId}	Get result for enrolment	Any (logged-in)
POST	/api/results	Record result	Organiser
GET	/api/participants/{id}/history	Get race history	Any (owner/organiser)

Common Conventions

URLs — All routes are lowercase, and resource names are plural nouns. Query parameters use camelCase (e.g., eventId).

Status codes — Standard REST statuses are used:

- 200 OK — success with body
- 201 Created — resource successfully created
- 204 No Content — successful delete with no body
- 400 Bad Request — validation error
- 401 Unauthorized — missing or invalid authentication
- 403 Forbidden — authenticated but lacks permission (wrong role or not owner)
- 404 Not Found — resource does not exist

This plan matches the entities and relationships defined in ERD.png and the schema in database-schema.sql.

Google Maps & Weather API Integration Plan

API endpoint plan for integrating Google Maps APIs and a Weather API. Covers Google's own Weather API (recommended for seamless integration) alongside popular third-party alternatives such as OpenWeatherMap and Weatherstack.

1. Google Maps Platform APIs

1.1 Routes API (recommended — replaces legacy Directions & Distance Matrix)

Function	Method	Endpoint	Description
Compute route	POST	https://routes.googleapis.com/directions/v2:computeRoutes	Calculates a route between two or more points, with real-time traffic support.
Compute distance matrix	POST	https://routes.googleapis.com/distanceMatrix/v2:computeRouteMatrix	Returns travel time and distance for multiple origins and destinations.

Legacy Directions API (not recommended): <https://maps.googleapis.com/maps/api/directions/json>

1.2 Geocoding API

Function	Method	Endpoint	Description
Forward geocoding	GET	https://maps.googleapis.com/maps/api/geocode/json?address=...	Converts an address to latitude/longitude.
Reverse geocoding	GET	https://maps.googleapis.com/maps/api/geocode/json?latlng=...	Converts coordinates to a human-readable address.

1.3 Places API (new version)

Function	Method	Endpoint	Description
Autocomplete	POST	https://places.googleapis.com/v1/places:autocomplete	Returns place predictions based on user input.
Place details	GET	https://places.googleapis.com/v1/places/{place_id}	Retrieves detailed information about a specific place.
Nearby search	POST	https://places.googleapis.com/v1/places:searchNearby	Searches for places near a given location.
Text search	POST	https://places.googleapis.com/v1/places:searchText	Searches places using a text query (e.g., "coffee shops in London").
Place photos	GET	https://places.googleapis.com/v1/places/*/photos/*/media	Serves photo media for a place.

1.4 Other Useful Google Maps APIs

API	Endpoint	Description
Elevation API	https://maps.googleapis.com/maps/api/elevation/json	Returns elevation data for a location.
Time Zone API	https://maps.googleapis.com/maps/api/timezone/json	Returns time zone information for a location.
Roads API	https://roads.googleapis.com/v1/snapToRoads	Snaps GPS tracks to the nearest road.
Maps Static API	https://maps.googleapis.com/maps/api/staticmap	Generates a static map image (PNG).

2. Weather API Endpoints

Option A — Google Weather API (recommended for Maps integration)

Google provides a native Weather API that uses the same authentication and billing as the Maps Platform.

Function	Method	Endpoint	Description
Current conditions	GET	https://weather.googleapis.com/v1/currentConditions:lookup	Returns current weather for a given location.
Daily forecast	GET	https://weather.googleapis.com/v1/forecast/days:lookup	Provides up to 10 days of daily forecasts.
Hourly forecast	GET	https://weather.googleapis.com/v1/forecast/hours:lookup	Provides up to 240 hours (10 days) of hourly forecasts.
Historical weather	GET	https://weather.googleapis.com/v1/history.hours:lookup	Returns historical weather data for a specific timestamp.

Option B — OpenWeatherMap (popular third party)

Function	Method	Endpoint	Description
Current weather	GET	https://api.openweathermap.org/data/2.5/weather	Fetches current weather by city name or coordinates.
5-day forecast	GET	https://api.openweathermap.org/data/2.5/forecast	5-day forecast with 3-hour intervals.
One Call API	GET	https://api.openweathermap.org/data/3.0/onecall	Combines current, hourly, and daily forecasts in one request.

Option C: Weatherstack

Function	Method	Endpoint	Description
Current weather	GET	http://api.weatherstack.com/current	Real-time weather for a specified location.
Historical weather	GET	http://api.weatherstack.com/historical	Weather for a specific date or date range.
Forecast	GET	http://api.weatherstack.com/forecast	Up to 14 days of daily forecast.
Location autocomplete	GET	http://api.weatherstack.com/locations	Searches for cities, regions, etc.

3. Authentication Methods

API	Authentication
Google Maps Platform APIs	API key (key=YOUR_API_KEY) or OAuth 2.0 (for certain endpoints).
Google Weather API	API key (shared with Maps).
OpenWeatherMap	API key (appid=YOUR_API_KEY).
Weatherstack	API key (access_key=YOUR_ACCESS_KEY).

4. Integration Suggestions

When using both Google Maps and a Weather API, typical workflows include:

- **Geocode an address → fetch weather.** Use the Geocoding API to get coordinates, then call the Weather API with those coordinates.
- **Plan a route → get weather along the route.** Use the Routes API to compute a route, extract key waypoints (e.g., every 50 km), and query weather for each.
- **Search for places → show their weather.** Use the Places API to find points of interest, then fetch current weather for each.